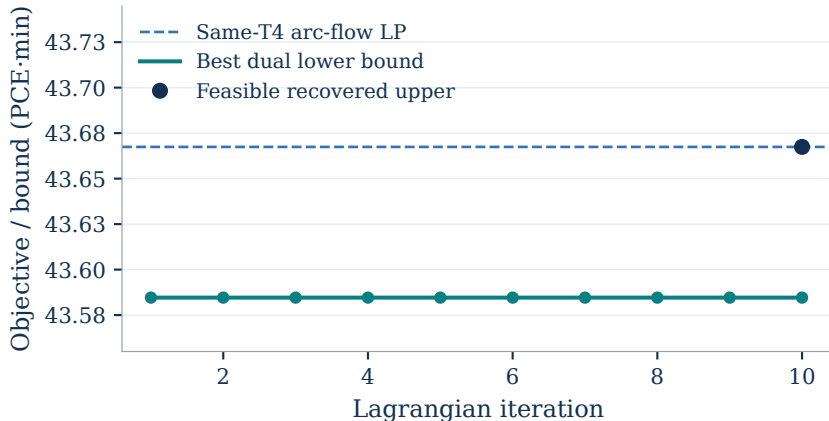
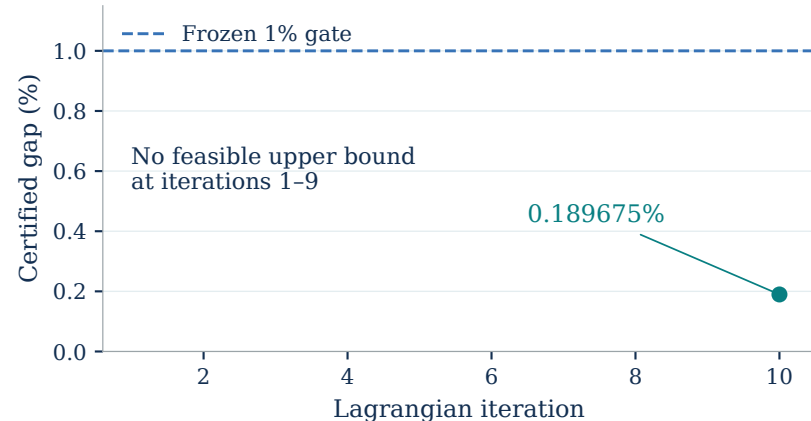


Lagrangian bounds and primal recovery

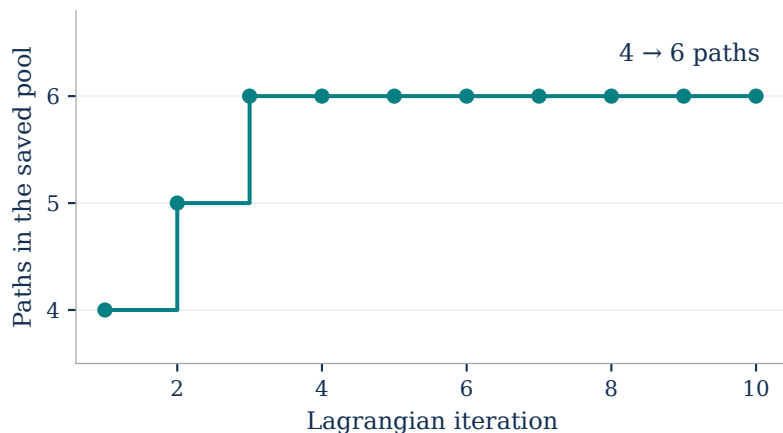
a Best bounds and same-T4 LP



b Available primal-dual certificate



c Saved path-pool growth



d Separate recovery LP calls

